

Quadratics in factored form



1

a

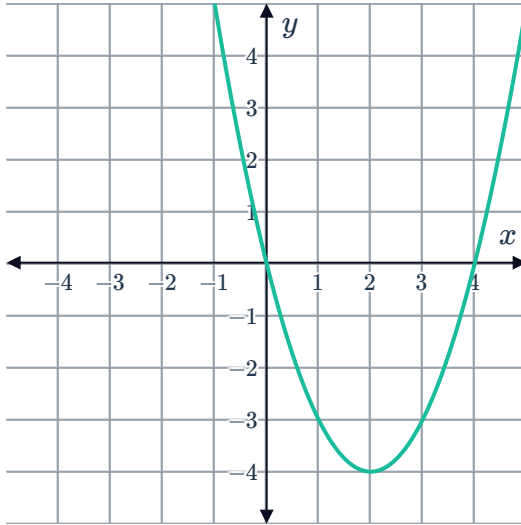
i $y = 0$

ii $x = 0, x = 4$

iii $x = 2$

iv $(2, -4)$

v



b

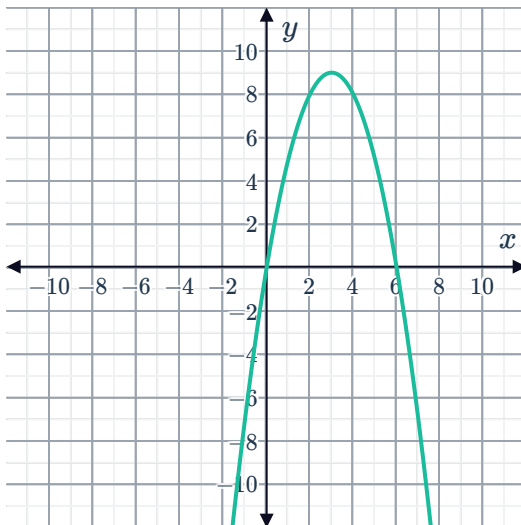
i $y = 0$

ii $x = 0, x = 6$

iii $x = 3$

iv $(3, 9)$

v

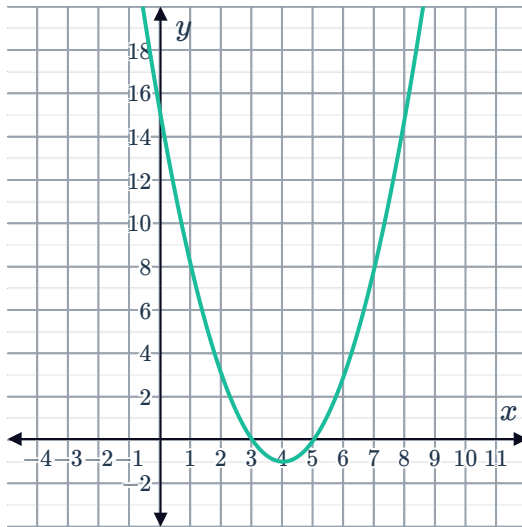


c

i $y = 15$

iii $x = 4$

v



ii $x = 5, x = 3$

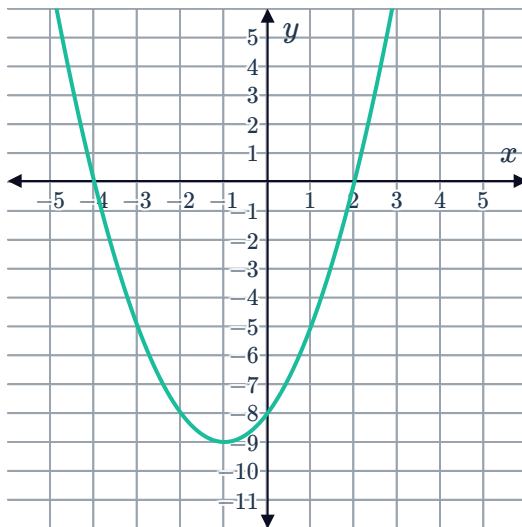
iv $(4, -1)$

d

i $y = -8$

iii $x = -1$

v



ii $x = -4, x = 2$

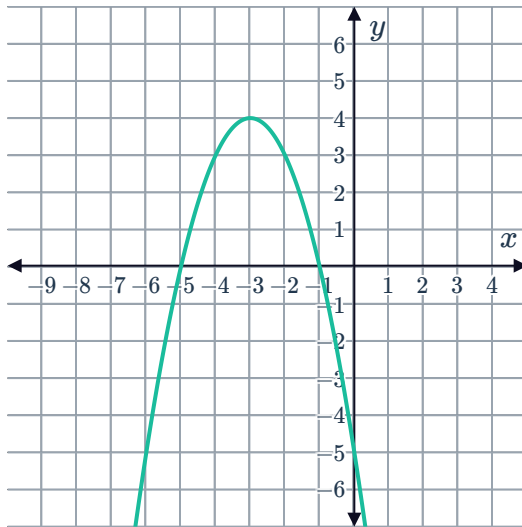
iv $(-1, -9)$

e

i $y = -5$

iii $x = 3$

v



ii $x = 1, x = 5$

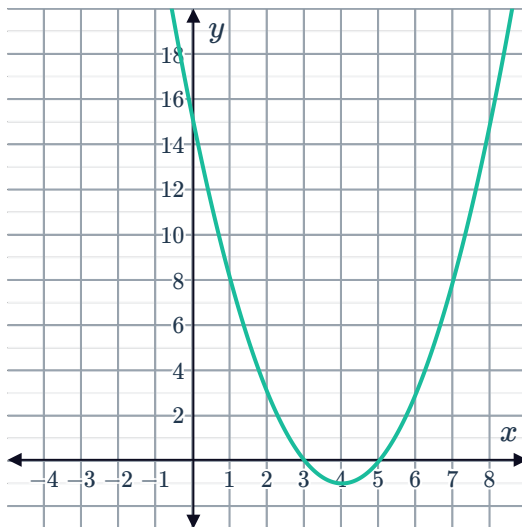
iv $(3, 4)$

f

i $y = 15$

iii $x = 4$

v



ii $x = 5, x = 3$

iv $(4, -1)$

2



a

i 0

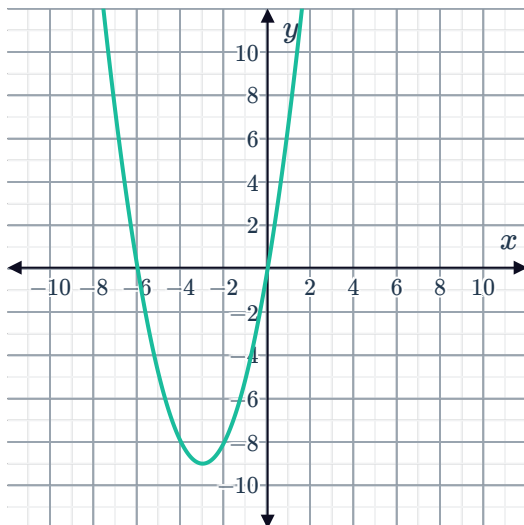
ii $x = 0, x = -6$

iii

iv $(-3, -9)$

x	-5	-4	-3	-2	-1
y	-5	-8	-9	-8	-5

v



b

i 12

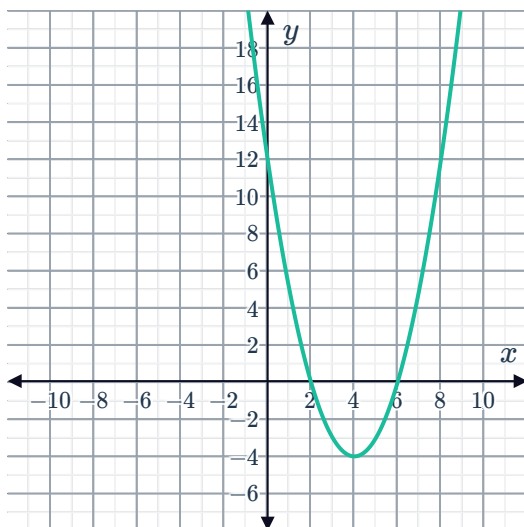
ii $x = 2, x = 6$

iii

iv $(4, -4)$

x	0	2	4	6	8
y	12	0	-4	0	12

v



c

i 8

iii

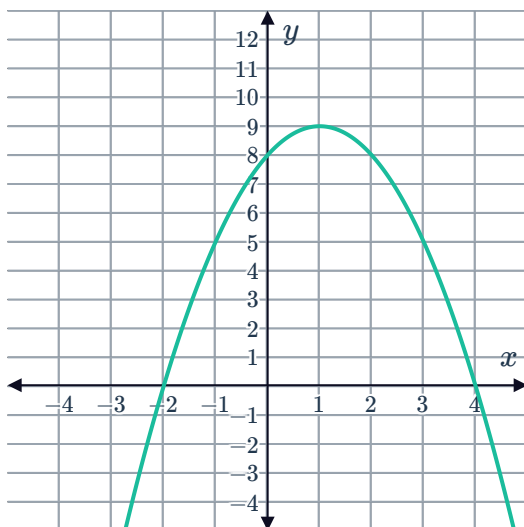
x	-5	-3	-1	1	3
y	-7	5	9	5	-7



ii $x = 2, x = -4$

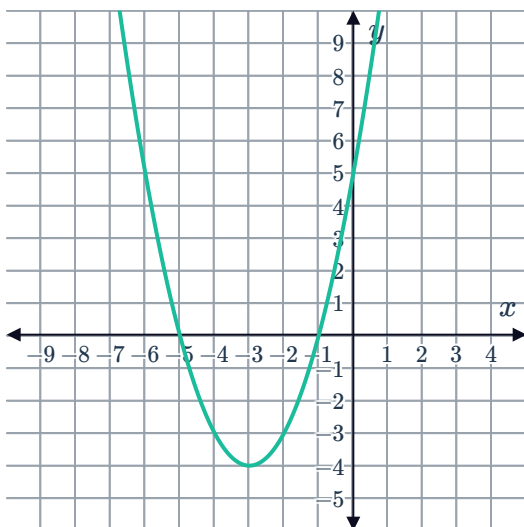
iv $(-1, 9)$

v



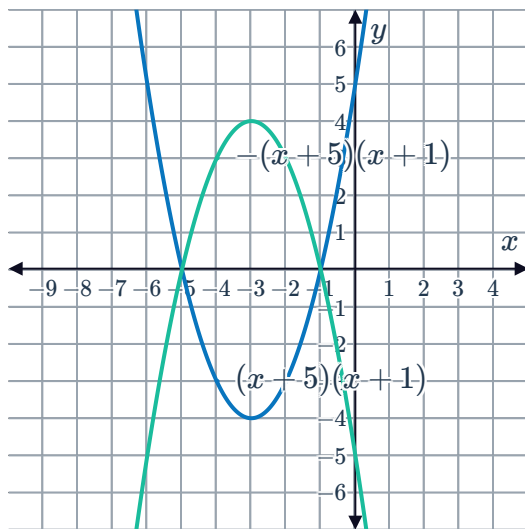
3

a



b -5

c



4

a B

b C

c D

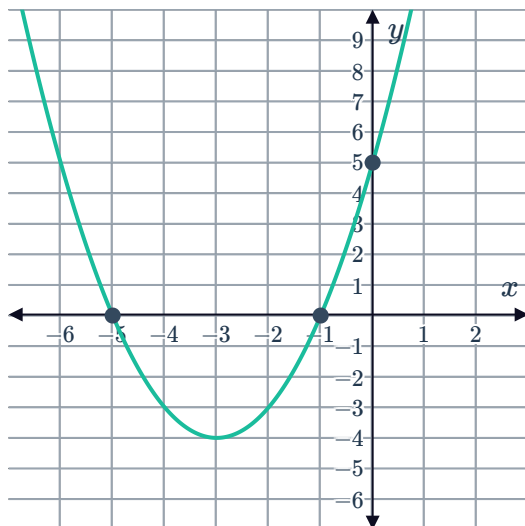
d A

5

a $y = (x + 1)(x + 5)$

b $y = 5$

c



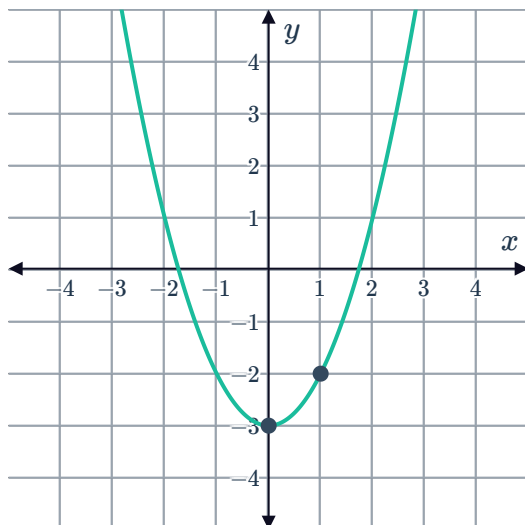
6

a $y = x^2 - 3$



b -2

c



7

a

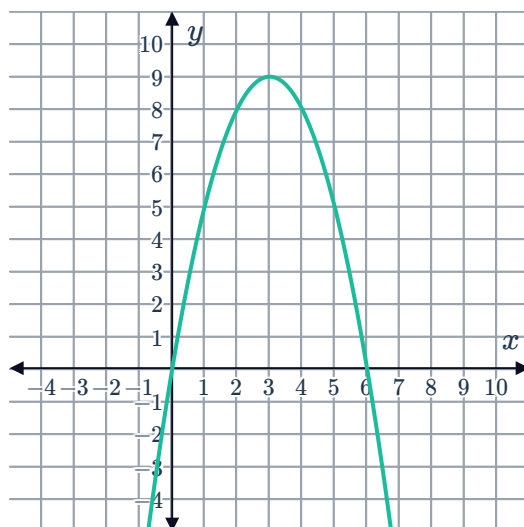
i $x(6 - x)$

ii $x = 0, x = 6$

iii 3

iv 9

v

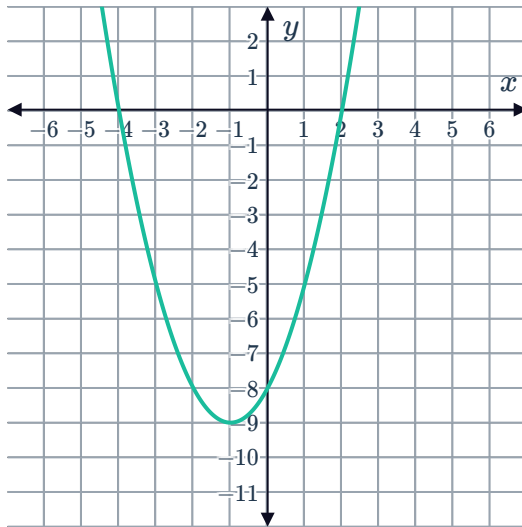


b

i $(x - 2)(x + 4)$

iii -1

v



ii $x = 2, x = -4$

iv -9

8

a $x(4 + x)$

c 0

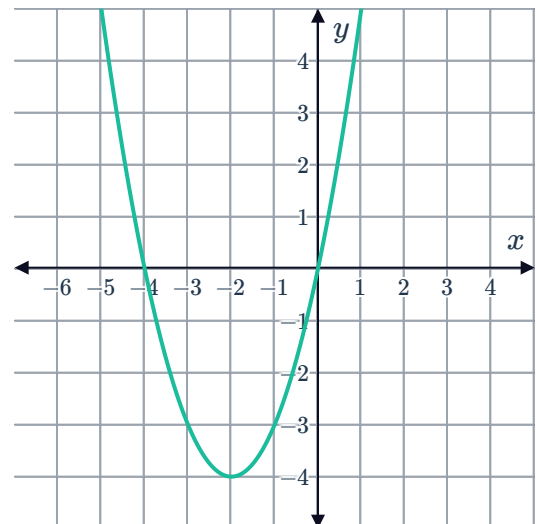
b $x = 0, x = -4$

d

x	-4	-3	-2	-1	0
y	0	-3	-4	-3	0

e $(-2, -4)$

f



9

a $(x + 2)(x + 6)$



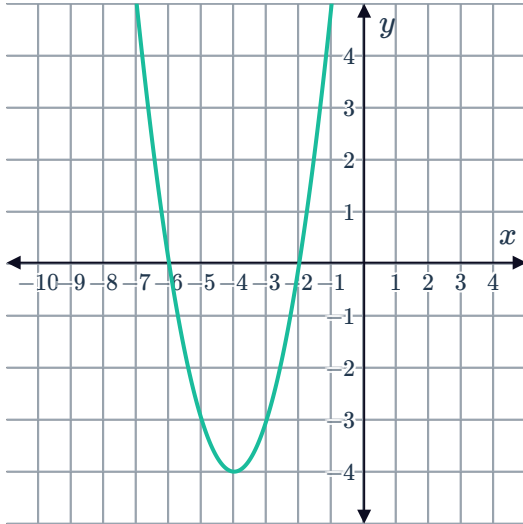
b $x = -2, x = -6$

c

x	-8	-6	-4	-2	0
y	12	0	-4	0	12

d $(-4, -4)$

e



10

a No

b Yes

11

a 8

b $y = -(x - 2)(x + 4)$

12 $x = 0$

13

a $x = -11$

b $y = 1(x - 1)(x + 11)$

c $(-5, -36)$

14

a $y = x^2 - b^2$

b $a = 0$

15

a $t = 20$

b 100 m



Additional questions

16 $y = a(x - x_1)(x - x_2)$

The values of x_1 and x_2 are the x -values of the x -intercepts of the quadratic function. The scale factor of the function is represented by a .

17 The axis of symmetry of a quadratic is the vertical line that can be drawn so each side of the quadratic function mirrors the other. This line passes through the vertex of the quadratic function.

18

a $x = 1$ and 4

b $x = 3$ and -7

c $x = 0$ and 4

d $x = 1$ and -5

19

a $y = 24$

b $y = 0$

20 $x = 1$

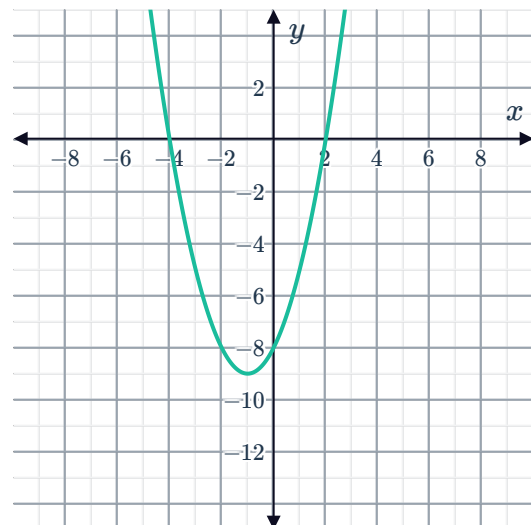
21

a i $(2, 0)$ and $(-4, 0)$

iii $(-1, -9)$

ii $x = -1$

iv

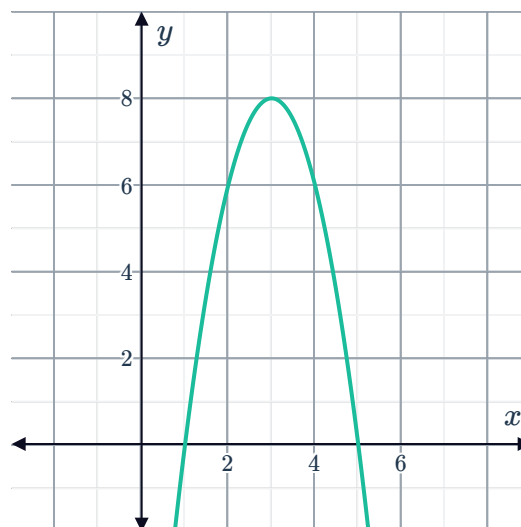


- b**
- i** (1, 0) and (5, 0)
 - iii** (3, 8)



ii $x = 3$

iv



22

- a**
- i** (1, 0) and (4, 0)
 - iii** $y = 2(x - 1)(x - 4)$
- b**
- i** (-2, 0) and (6, 0)
 - iii** $y = -\frac{1}{3}(x - 6)(x + 2)$
- ii** (0, 8)
- ii** (0, 4)

23 $y = -4(x - 5)(x + 3)$

24 $y = \frac{1}{3}(x + 2)(x - 9)$

25 $y = 3(x + 4)(x + 7)$

26

- a** (50, 110)
- b** $y = -\frac{11}{250}x(x - 100)$
- c** Domain: $0 \leq x \leq 100$
Range: $0 \leq y \leq 110$

27

- a** (20, 0) and (40, 0)
- b** $y = -\frac{1}{20}(x - 20)(x - 40)$

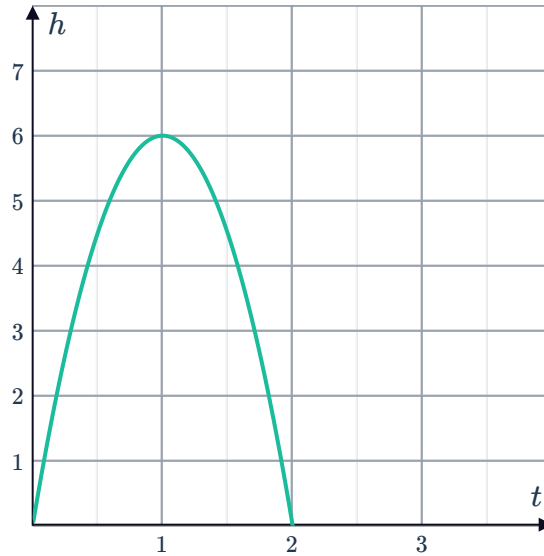


28

a The x -intercepts represent the times when the eraser is 0 ft above Wilson's hand. This will be exactly when Wilson tosses the eraser and when it returns to his hand. Since Wilson tosses the eraser at $x = 0$ and it takes 2 seconds for it to return to his hand, the x -intercepts are $(0, 0)$ and $(2, 0)$. Since one of the x -intercepts is at the origin, we have the y -intercept is also $(0, 0)$.

b $(1, 6)$

c



d $y = -6x(x - 2)$

29 Since the vertex lies on the axis of symmetry, we know that 8 is the average of 12 and x_0 . So then, we must have $x_0 = 4$.

If we substitute the coordinates of the vertex into the function, we get:

$$15 = k(8 - 4)(8 - 12)$$

If we solve this equation we get $k = -\frac{15}{16}$.